LEARNER GUIDE

Equations with fractions

Equations, Inequalities and Simultaneous Relationships · Lesson 3. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A fraction like $x/3$ just means x shared into 3 equal parts. Because 3 lots of $x/3$ rebuild the whole x, the way to handle a fraction in an equation is to **multiply every term by the denominator** (the number underneath) — the fraction disappears, and you are left with an equation you already know how to solve. If two different denominators appear, multiply by their **lowest common multiple** (the smallest number both divide into) to clear them both at once. As always, you **check** by substituting your answer back.

What you are learning to notice

- x/n means one of n equal parts; n lots of x/n make a whole x.
- To clear x/n, multiply by n — and multiply **every** term, both sides, to keep it true.
- With two denominators, multiply by their **lowest common multiple** to clear both at once.
- After clearing, it is an ordinary equation — and the substitution **check** still proves the answer.

Moving through it, one step at a time

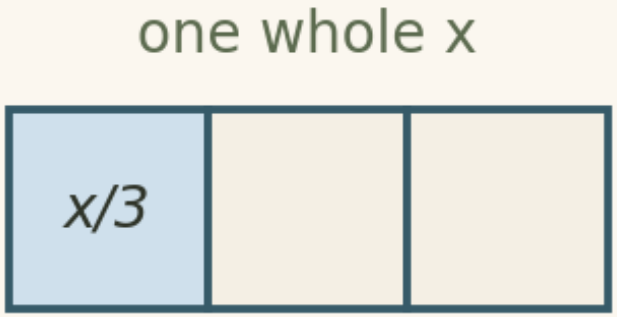
First clear the fraction. Multiply every term on both sides by the denominator (or, with two denominators, by their lowest common multiple). Take care to multiply *every* term — miss one and the equation is no longer true. Now you have a familiar equation: expand or collect if needed, clear the number by adding or subtracting on both sides, then divide. Finally substitute your answer into the original and check both sides match.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — A single fraction

Solve $x/3 + 2 = 5$. The bar shows $x/3$ — one of three equal parts of x.

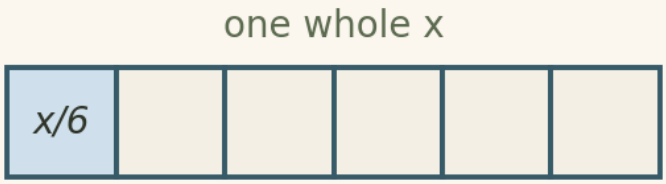


1. Multiply every term by 3 to clear the fraction.	$3\times(x/3) + 3\times2 = 3\times5$
2. The fraction is gone.	$x + 6 = 15$
3. Subtract 6 from both sides.	$x = 9$
4. Check in the original.	$9/3 + 2 = 3 + 2 = 5$ ✓

$x = 9$ — multiply every term by 3 to get $x + 6 = 15$, then subtract 6; the check gives 5.

Example 2 — Two denominators — use the LCM

Solve $x/2 + x/3 = 10$. The denominators are 2 and 3; their lowest common multiple is 6.



1. Multiply every term by 6 (the LCM of 2 and 3).	$6\times(x/2) + 6\times(x/3) = 6\times10$
2. Both fractions clear at once.	$3x + 2x = 60$
3. Collect like terms.	$5x = 60$
4. Divide both sides by 5.	$x = 12$
5. Check in the original.	$12/2 + 12/3 = 6 + 4 = 10$ ✓

$x = 12$ — the LCM of 2 and 3 is 6; multiplying through gives $3x + 2x = 60$, so $5x = 60$ and $x = 12$.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (fractions of a quantity, the lowest common multiple, and keeping an equation balanced), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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